

INPPL1 fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes INPPL1 gene fusions from the msk_impact_50k_2026 study, covering 14 fusion events. 4/14 fusions (28.6%) were in-frame. 7/14 fusions (50.0%) retained the Endonuclease/Exonuclease/phosphatase family domain. Domain retention was tested with Fisher's exact test ($p=0.048951$, statistically significant at $\alpha=0.05$). No literature benchmark is configured for INPPL1.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 14 analyzed fusion events, identifying 11 distinct fusion partner genes. The most frequent partner was FCHSD2, observed in 4 events.

Domain retention

Endonuclease/Exonuclease/phosphatase family domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 4/4 (100.0%) of in-frame fusions retained the domain, $p=0.048951$ (statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.108911.

Domain disruption

Domain-disruption analysis was skipped: INPPL1 has no disruption_required_domains configured; domain-disruption analysis was skipped.

Cutpoint detection

Cutpoint detection scanned 13 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 560 aa (permutation-corrected $p=0.033966$, statistically significant at $\alpha=0.05$). This is 131 aa from the nearest configured domain boundary at 429 aa.

Corroborating confidence statistics

Corroborating confidence statistics were not computed for this run: INPPL1 has no group_field configured; confidence-stats analysis was skipped.

Composite evidence score

Composite evidence score reported: domain_disruption sub-score excluded: the supplied result had no applicable p-value (e.g. disruption_required_domains not configured for this gene).

Exon retention

Exon retention reported: No target exon was supplied in params or GeneConfig.

Joint partner

Joint partner reported: INPPL1 has no gene_pair configured; joint-partner analysis was skipped.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
ANK2	1	0.07142857142857142	0.09629286886244175	None	1.0	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2792882694916884	1
SHANK2	1	0.07142857142857142	0.09629286886244175	None	0.956	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.26985969806311694	2
FCHSD2	4	0.2857142857142857	0.09629286886244175	None	0.51	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2661250041855659	3
PTCH1	1	0.07142857142857142	0.09629286886244175	None	0.608	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.1952882694916884	4
GAB2	1	0.07142857142857142	0.09629286886244175	None	0.31999999999999995	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.13357398377740265	5
NLRP6	1	0.07142857142857142	0.09629286886244175	None	0.29200000000000004	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.12757398377740267	6
CLPB	1	0.07142857142857142	0.09629286886244175	None	0.246	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.11771684092025979	7
FOLR2	1	0.07142857142857142	0.09629286886244175	None	0.14600000000000002	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.09628826949168838	8
FAT3	1	0.07142857142857142	0.09629286886244175	None	0.10799999999999998	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.08814541234883122	9
SLC67A1	1	0.07142857142857142	0.09629286886244175	None	None	None	['domain_retention', 'recurrence']	0.08273052480760337	10
LRTOMT	1	0.07142857142857142	0.09629286886244175	None	0.0	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.06500255520597409	11

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
114	0	7	1	5	0.0	0.46153846153846156	0.3357921019231931

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
133	0	7	2	4	0.0	0.1923076923076923	0.7160033436347992
183	1	6	2	4	0.3333333333333333	0.5594405594405595	0.2522460504731182
206	1	6	3	3	0.16666666666666666	0.26573426573426573	0.5755524408482516
220	1	6	4	2	0.08333333333333333	0.10256410256410256	0.9890046156985368
364	2	5	4	2	0.2	0.2861305361305362	0.5434357913897181
538	2	5	5	1	0.08	0.10256410256410256	0.9890046156985368
560	2	5	6	0	0.0	0.020979020979020976	1.6782147827453995
805	6	1	6	0	0.0	1.0	-0.0

domain_retention tables

frame_domain_contingency_table

4	3
0	6

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
auto_key_domain	Endonuclease/Exonuclease/phosphatase family	13	7	2	4	0.17988505747126438

frequency tables

Partner_gene_counts

Partner_gene	Event_count
ANK2	1
CLPB	1
FAT3	1
FCHSD2	4
FOLR2	1
GAB2	1
LRTOMT	1
NLRP6	1
PTCH1	1
SHANK2	1

Partner_gene	Event_count
SLC67A1	1

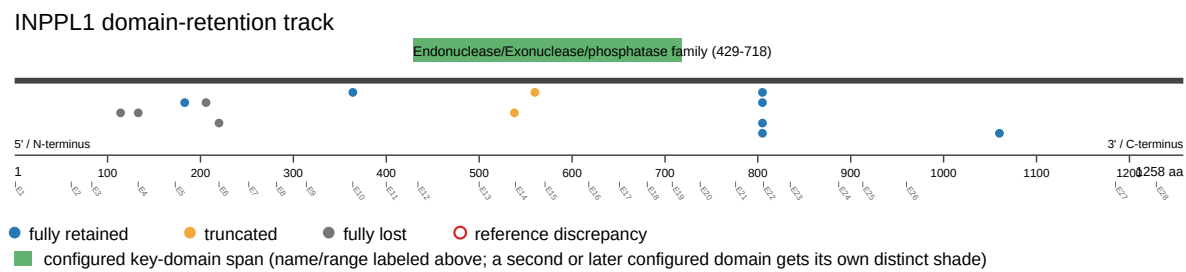
Per-event results (results.tsv)

event_id	sample_id	patient_id	fusion_name	partner_gene	frameshift	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domains	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0007343-T02-IM6-4	P-0007343-T02-IM6		INPPL1-ANK2	ANK2	unknown	five_prime	71943348	14	560	False	disrupted	0.45517241379310347	True			Endonuclease/Exonuclease/phosphatase family	INPPL1 (NM_001567) - ANK2 (NM_001148) rearrangement: t(4;11)(q25-q26;q13.4)(chr4:g.113898703::chr11:g.71943348) Protein Fusion: mid-exon {INPPL1:ANK2}	NA
EVT-P-0031523-T02-IM6-5	P-0031523-T02-IM6		INPPL1-CLPB	CLPB	unknown	three_prime	71940164	5	183	False	retained	1.0	False	Endonuclease/Exonuclease/phosphatase family			INPPL1 (NM_00156) rearrangement: c.549:INPPL1_chr11:g.72024501del Antisense Fusion	NA
EVT-P-0042110-T02-IM7-6	P-0042110-T02-IM7		INPPL1-FAT3	FAT3	unknown	five_prime	71939487	3	114	False	lost	0.0	False		Endonuclease/Exonuclease/phosphatase family		INPPL1 (NM_001567) - FAT3 (NM_001008781) rearrangement: c.342:INPPL1_c.3669+26772: FAT3del Protein Fusion: mid-exon {INPPL1:FAT3}	NA
EVT-P-0004226-T08-IM7-7	P-0004226-T08-IM7		INPPL1-FCHSD2	FCHSD2	in-frame	five_prime	71945806	21	805	True	retained	1.0	False	Endonuclease/Exonuclease/phosphatase family			INPPL1 (NM_001567) - FCHSD2 (NM_014824) rearrangement: c.2415+147:INPPL1_c.705+16442:FCHSD2inv Protein Fusion: in frame {INPPL1:FCHSD2}	NA
EVT-P-0004226-T04-IM6-8	P-0004226-T04-IM6		INPPL1-FCHSD2	FCHSD2	in-frame	five_prime	71945806	21	805	True	retained	1.0	False	Endonuclease/Exonuclease/phosphatase family			INPPL1 (NM_001567) - FCHSD2 (NM_014824) rearrangement: c.2415+147:INPPL1_c.705+16442:FCHSD2inv Protein Fusion: in frame {INPPL1:FCHSD2}	NA
EVT-P-0004226-T03-IM6-9	P-0004226-T03-IM6		INPPL1-FCHSD2	FCHSD2	in-frame	five_prime	71945742	21	805	True	retained	1.0	False	Endonuclease/Exonuclease/phosphatase family			INPPL1 (NM_001567) - FCHSD2 (NM_014824) rearrangement: c.2415+83:INPPL1_c.705+16440:FCHSD2inv Protein Fusion: in frame {INPPL1:FCHSD2}	NA

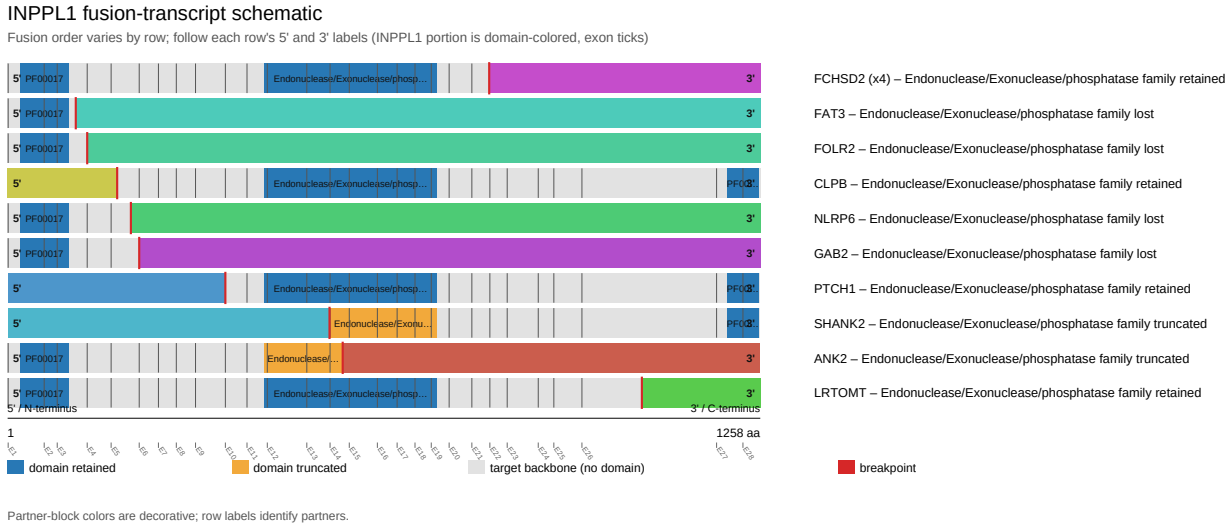
event_id	samp le_id	p a t i e n t _i d	fusio n_n ame	par tne r_g en e	fra me _st atu s	ta rg et _r ole	break point_ geno mic	brea kpoi nt_e xon	breakpoi nt_protei n_positio n	is_intr onic_b reakpoi nt	do mai n_s tatu s	domain_ retained_ _fraction	domai n_is_t runcat ed	retained_dom ains	lost_domains	disrupted_do mains	source_annotation_te xt	source_sit e2_effec t_on_frame
EVT-P-0 004226-T 07-IM7-1 0	P-000 4226- T07-I M7		INP PL1- FCH SD2	FC HS D2	in-f ra me	fiv e_ pri me	71945 792	21	805	True	reta ined	1.0	False	Endonuclease/ Exonuclease/p hosphatase family			INPPL1 (NM_001567) - FCHSD2 (NM_014824) rearrangement: c.2415+ 133:INPPL1_c.705+164 41:FCHSD2inv Protein Fusion: in frame {INPPL1:FCHSD2}	NA
EVT-P-0 045578-T 01-IM6-1 1	P-004 5578- T01-I M6		INP PL1- FOL R2	FO LR 2	out -of- fra me	fiv e_ pri me	71939 641	3	133	True	lost	0.0	False		Endonuclease/ Exonuclease/p hosphatase family		INPPL1 (NM_001567) - FOLR2 (NM_001113536) rearrangement: c.397+9 9:INPPL1_c.151-191:F OLR2dup Protein Fusion: out of frame {INPPL1:FOLR2}	NA
EVT-P-0 009791-T 02-IM6-1 2	P-000 9791- T02-I M6		INP PL1- GAB 2	GA B2	out -of- fra me	fiv e_ pri me	71940 493	6	220	True	lost	0.0	False		Endonuclease/ Exonuclease/p hosphatase family		INPPL1 (NM_001567) - GAB2 (NM_080491) rearrangement: c.660-1 6:INPPL1_c.75+18325: GAB2inv Protein Fusion: out of frame {INPPL1:GAB2}	NA
EVT-P-0 048151-T 01-IM6-3 1	P-004 8151- T01-I M6		INP PL1- LRT OMT	LR TO MT	un kn ow n	fiv e_ pri me	71948 466	26	1060	False	reta ined	1.0	False	Endonuclease/ Exonuclease/p hosphatase family			INPPL1 (NM_001567) - LRTOMT (NM_145309) rearrangement: c.3178:I NPPL1_c.83-1241:LRT OMTdup Protein Fusion: mid-exon {INPPL1:LRTOMT}	NA
EVT-P-0 005693-T 03-IM6-3 4	P-000 5693- T03-I M6		INP PL1- PTC H1	PT CH 1	out -of- fra me	thr ee_ p ri me	71941 318	9	364	True	reta ined	1.0	False	Endonuclease/ Exonuclease/p hosphatase family			PTCH1 (NM_000264) - INPPL1 (NM_001567) rearrangement: t(9;11)(q22.32;q13.4)(chr9:g.98 211830::chr11:g.71941 318) Protein Fusion: out of frame {PTCH1:INPPL1}	NA
EVT-P-0 024858-T 01-IM6-3 6	P-002 4858- T01-I M6		NLR P6-I NPP L1	NL RP 6	un kn ow n	fiv e_ pri me	71940 232	5	206	False	lost	0.0	False		Endonuclease/ Exonuclease/p hosphatase family		INPPL1 (NM_001567) - NLRP6 (NM_138329) rearrangement: c.617:I NPPL1_c.2138:NLRP6 dup Protein Fusion: mid-exon {INPPL1:NLRP6}	NA
EVT-P-0 031882-T 01-IM6-3 7	P-003 1882- T01-I M6		SHA NK2- INP PL1	SH AN K2	un kn ow n	thr ee_ p ri me	71942 656	13	538	False	disr upt ed	0.624137 9310344 828	True			Endonuclease/ Exonuclease/p hosphatase family	INPPL1 (NM_001567) rearrangement: c.1612:I NPPL1_chr11:g.705015 15dup Antisense Fusion	NA

Figures

domain retention outliers



fusion schematic



reference comparison

Reference vs msk_impact_50k_2026

